

MIS 512: DATA MANAGEMENT

Group Project

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STEP 1) REAL BUSINESS

Riyadh Air

STEP 2) BACKGROUD INFORMATION

Riyadh Air is a newly established airways announced as part of Saudi vision 2030 to encompass the latest and largest fleet of airplanes to service the region and become one of the largest airlines globally. The company is based in Riyadh being its operational hub and is planned to avail over 200,000 direct and in-direct jobs. The current fleet of airplanes is currently standardized to the latest Boeing 787-9 which will serve the vision of making Riyadh a hub connecting the east and west hemisphere with over 100 global destinations. It is imperative that Riyadh Air builds a profound data base system that serves the vision intended.

STEP 3 & STEP 4) INVESTIGATION AND BUSINESS PROBLEM IDENTIFIED

As a new entity Riyadh Air needs to establish a state-of-the-art database management system that will handle the forecasted large operations. The database should be flexible and well-constructed to allow the future expansion of its airplane fleet and staff.

The database's main objective is to handle the daily operations of the airline and manage customer reservations and staff assignments. The database will record the daily flight schedules including the airplane type, cabin type, cabin fair, date, time, route. The pilots assigned to the airplane require to be identified and the name, ID, Email, pilot license number and type. Each customer booking a flight will have his/her name recorded along with government or passport ID, phone number, email, date of birth and nationality. The reservation requires you to have a unique ID with details of cabin type, number of luggage, ticket number, date of purchase.

STEP 5) BUSINESS RULES

The following relational Database keeps track of RiyadhAir Business activities. Due to high reliability on the company, causing day to day expanding business activities hard to manage its operations, we are developing a database for its airline reservation system, a computerized system to ensure more flexible service for customers that will help manage all accounting information related to flight operations, passengers, reservation number, schedule publishing airfare payments etc. The database should record data about its Passengers, Flight, Reservation, Cabins, Pilot, Ticket, Airplane

1. Pilot:

- Attributes: Pilot ID (Primary Key), PFName, PLName, Contact Number, Pilot License.
- Relationships: Each flight has one and only one pilot, yet each pilot can be assigned to multiple flights

2. Flights:

- Attributes: FlightID (Primary Key), Flight_Fare, Destination, Departure_From, Arrival_Date, Departure_Date

- Relationships: Each flight has one pilot, each flight has multiple seats, each flight has multiple reservations.

3. Cabin_Type:

- Attributes: Cabin_Type(Primary Key), Cabin Fare
- Relationships: Each cabin will be assigned to one and only one airplane, each cabin can have multiple seats.

4. Reservation:

- Attributes: Reservation_ID (Primary Key), Date_of_Departure, Date_of_Arrival, Luggage_Allowance
- Relationships; Each reservation will have one and only one flight, each reservation can have multiple tickets

5. Airplane:

- Attributes: Airplane_ID (Primary Key), Modelnumber, Capacity
- Relationships: Each airplane can be used for many flights.

6. Passengers:

- Attributes: Passenger_ID (Primary Key), FName, LName, Email, Phone_Num, Nationality, DOB, Gender.
- Relationships: Each passenger can have only one reservation

7. Ticket:

- Attributes: Ticket_ID (Primary Key), Fare, Meal
- **Relationships:** Each ticket has one reservation

- NOTE

The base air fare is defined for the flight where the fare of each cabin is derived from

Economy Cabin Fare = Flight Base Fare + 5%

Business Cabin Fare = Flight Base Fare + 75%

First Cabin Fare = Flight Base Fare + 150%

STEP 6.A) DATABASE DESIGN

- **PILOT** (Pilot_ID (PK), PName, PLName, Contact_Number, Pilot License)
- **FLIGHT** (Flight_ID, Flight_Fare, Departure_From, Destination, Arrival_Date, Departure_Date, Pilot_ID(FK1))
- **RESERVATION** (ReservationID (PK), Date_of_Departure, Date_of_Arrival, Luggage_Allowance, Flight_ID(FK1))
- **AIRPLANE** (AirplaneID (PK), Model Number, Capacity)
- **PASSENGER** (Passenger_ID (PK), FName, LName, Email, Phone Num, Nationality, DOB, Gender, Reservation_ID(FK1))
- **TICKET** (Ticket_ID(PK), Reservation_ID (FK1), Fare, Meal)
- **CABIN_TYPE** (Cabin_Type (PK), Airplane_ID (PK, FK1), Cabin_Fare)
- **SEAT** (Seat_ID (PK), Flight_ID (FK1), Cabin_Type (FK2), Airplane_ID (FK2))

STEP 6.B) THE RELATIONSHIP ARE AS FOLLOWS:

Passenger and Reservation:

- Each passenger can have many reservations, but only one active reservation at a time.
- Each reservation can belong to one or many passengers.
- Reservations can be modified depending on availability and within a specified timeframe.

Passenger and Ticket:

- Each passenger can have only one ticket issued per reservation.
- Each ticket can belong to only one passenger.
- Tickets are issued based on confirmed reservations.

Seat and Cabin Type:

- Each seat has a single cabin type.
- Each cabin type could exist on many Seats.
- The cabin type determines the cabin fare, which is included in calculating the ticket fare.

Flight and Base Air Fare:

- The base air fare is defined for each flight.
- The fares for each cabin type are derived from the base air fare as follows:
 - Economy Cabin Fare = Base Air Fare + 5%
 - Business Cabin Fare = Base Air Fare + 75%
 - First Cabin Fare = Base Air Fare + 150%

Flight and Ticket:

- A ticket cannot be assigned to multiple flights.
- The number of tickets issued for a flight cannot exceed the airplane's capacity.

Flight and Airplane:

- Each flight is operated by a single airplane.
- Each airplane can be used for many flights.

- The airplane's capacity determines the maximum number of passengers allowed on a flight.

Flight and Seat:

- Each flight has many seats.
- The number of available seats on a flight is determined by the airplane's capacity and any existing bookings.

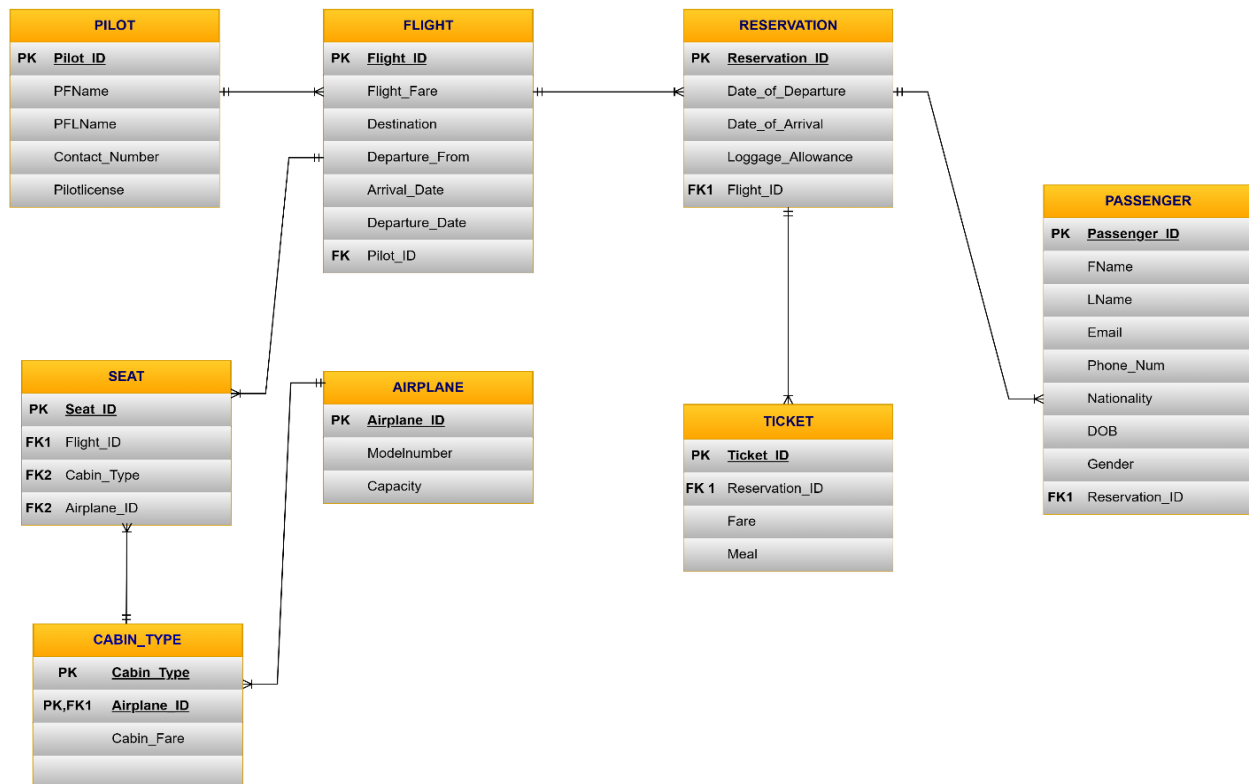
Pilot and Flight:

- Each flight can have one and only pilots.
- Each pilot can be assigned to many flights.
- Departure date and arrival date should not contradict another flight for the same pilot.

General Business Rules:

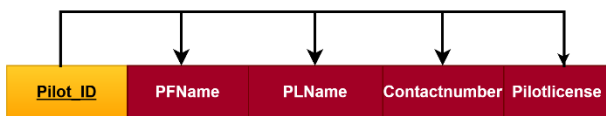
- All passengers must have unique identifiers and identifiable information.
- All flights and tickets must have unique identifiers for tracking purposes.
- Dates for departure, arrival, and reservation should be valid and consistent.
- ticket fares should be consistent with the cabin type and the flight fare.

STEP 7) Entity Relationship Diagram:



STEP 8) Normalization

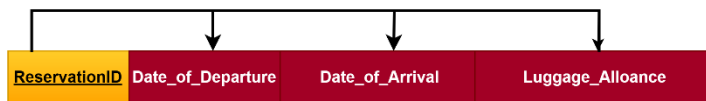
1) Table name: PILOT



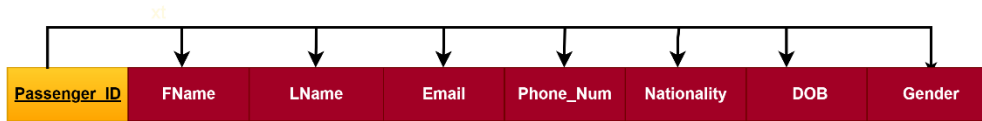
2) Table name: FLIGHT



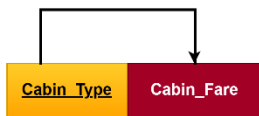
3) Table name: RESERVATION



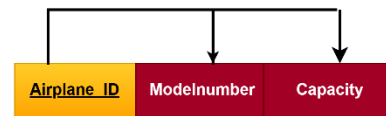
4) Table name: PASSENGER



5) Table name: CABIN_Type



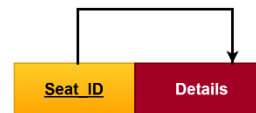
6) Table name: AIRPLANE



7) Table name: TICKET



8) Table name: SEAT



STEP 9.A) CREATE DATABASE TABLES

-- Create the AIRPLANE table

```
CREATE TABLE AIRPLANE (
    AIRPLANE_ID VARCHAR(10) NOT NULL PRIMARY KEY,
    MODEL_NUMBER VARCHAR(50) NOT NULL,
    CAPACITY INT NOT NULL
);
```

-- Create the PILOT table

```
CREATE TABLE PILOT (
    PILOT_ID VARCHAR(10) NOT NULL PRIMARY KEY,
    PFNAME VARCHAR(50) NOT NULL,
```

```
PLNAME VARCHAR(50) NOT NULL,  
CONTACT_NUMBER VARCHAR(20) NOT NULL,  
PILOT_LICENSE VARCHAR(50) NOT NULL  
);
```

-- Create the FLIGHT table

```
CREATE TABLE FLIGHT (  
    FLIGHT_ID VARCHAR(10) NOT NULL PRIMARY KEY,  
    FLIGHT_FARE DECIMAL(10,2) NOT NULL,  
    DESTINATION VARCHAR(50) NOT NULL,  
    DEPARTURE_FROM VARCHAR(50) NOT NULL,  
    ARRIVAL_DATE DATE NOT NULL,  
    DEPARTURE_DATE DATE NOT NULL,  
    PILOT_ID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (PILOT_ID) REFERENCES PILOT(PILOT_ID)  
);
```

-- Create the RESERVATION table

```
CREATE TABLE RESERVATION (  
    RESERVATION_ID VARCHAR(10) NOT NULL PRIMARY KEY,  
    DATE_OF_DEPARTURE DATE NOT NULL,  
    DATE_OF_ARRIVAL DATE NOT NULL,  
    LUGGAGE_ALLOWANCE INT NOT NULL,  
    FLIGHT_ID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (FLIGHT_ID) REFERENCES FLIGHT(FLIGHT_ID)  
);
```

-- Create the TICKET table

```
CREATE TABLE TICKET (  
    TICKET_ID VARCHAR(10) NOT NULL PRIMARY KEY,  
    RESERVATION_ID VARCHAR(10) NOT NULL,  
    FARE DECIMAL(10,2) NOT NULL,  
    MEAL VARCHAR(30) NOT NULL,  
    FOREIGN KEY (RESERVATION_ID) REFERENCES RESERVATION(RESERVATION_ID)  
);
```

-- Create the PASSENGER table

```
CREATE TABLE PASSENGER (  
    PASSENGER_ID VARCHAR(10) NOT NULL PRIMARY KEY,  
    FNAME VARCHAR(50) NOT NULL,  
    LNAME VARCHAR(50) NOT NULL,  
    EMAIL VARCHAR(100) NOT NULL UNIQUE,  
    PHONE_NUM VARCHAR(20) NOT NULL,  
    NATIONALITY VARCHAR(50) NOT NULL,  
    DOB DATE NOT NULL,  
    GENDER VARCHAR(6) NOT NULL,  
    RESERVATION_ID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (RESERVATION_ID) REFERENCES RESERVATION(RESERVATION_ID)  
);
```

-- Create the CABIN_TYPE table

```
CREATE TABLE CABIN_TYPE (  
    CABIN_TYPE VARCHAR(20) NOT NULL,  
    CABIN_FARE DECIMAL(10,2) NOT NULL,
```

```
AIRPLANE_ID VARCHAR(10) NOT NULL,  
PRIMARY KEY (CABIN_TYPE, AIRPLANE_ID),  
FOREIGN KEY (AIRPLANE_ID) REFERENCES AIRPLANE(AIRPLANE_ID)  
);
```

```
CREATE TABLE SEAT (  
    SEAT_ID VARCHAR(10) NOT NULL PRIMARY KEY,  
    CABIN_TYPE VARCHAR(20) NOT NULL,  
    AIRPLANE_ID VARCHAR(10) NOT NULL,  
    FLIGHT_ID VARCHAR(10) NOT NULL,  
    FOREIGN KEY (CABIN_TYPE, AIRPLANE_ID) REFERENCES CABIN_TYPE(CABIN_TYPE,  
AIRPLANE_ID),  
    FOREIGN KEY (FLIGHT_ID) REFERENCES FLIGHT(FLIGHT_ID)  
);
```

STEP 9.B) USING SQL CREATE DEMO DATA

```
INSERT INTO AIRPLANE (AIRPLANE_ID, MODEL_NUMBER, CAPACITY)  
VALUES ('AIRPLANE01', 'Airbus A320', 180),  
('AIRPLANE02', 'Boeing 737-800', 189),  
('AIRPLANE03', 'Airbus A350-900', 325),  
('AIRPLANE04', 'Boeing 787-9', 312),  
('AIRPLANE05', 'Airbus A380-800', 544),
```

```
('AIRPLANE06', 'Boeing 747-8', 467),  
('AIRPLANE07', 'Airbus A220-300', 145),  
('AIRPLANE08', 'Embraer 195-E2', 146),  
('AIRPLANE09', 'ATR 72-600', 72),  
('AIRPLANE10', 'De Havilland Canada DHC-8-400', 86);
```

```
INSERT INTO PILOT (PILOT_ID, PFNAME, PLNAME, CONTACT_NUMBER, PILOT_LICENSE)  
VALUES ('PILOT01', 'Ahmed', 'Ali', '+966 50 1234567', '123456789'),  
('PILOT02', 'Fatima', 'Mohammed', '+966 51 7654321', '987654321'),  
('PILOT03', 'Omar', 'Abdullah', '+966 55 9876543', '234567890'),  
('PILOT04', 'Sarah', 'Ahmed', '+966 54 3210987', '098765432'),  
('PILOT05', 'Khalid', 'Hasan', '+966 56 7890123', '345678901'),  
('PILOT06', 'Amira', 'Omar', '+966 53 1234987', '456789012'),  
('PILOT07', 'Hassan', 'Ali', '+966 52 9870123', '567890123'),  
('PILOT08', 'Halima', 'Fatima', '+966 57 0123456', '678901234'),  
('PILOT09', 'Mohammed', 'Khalid', '+966 58 4567890', '789012345'),  
('PILOT10', 'Aisha', 'Sarah', '+966 59 8901234', '890123456');
```

```
INSERT INTO FLIGHT (FLIGHT_ID, FLIGHT_FARE, DESTINATION, DEPARTURE_FROM,  
ARRIVAL_DATE, DEPARTURE_DATE, PILOT_ID)  
VALUES ('FLIGHT01', 1000, 'Riyadh', 'Jeddah', '2023-12-10', '2023-12-10', 'PILOT01'),  
('FLIGHT02', 1100, 'Jeddah', 'Dammam', '2023-12-11', '2023-12-11', 'PILOT02'),  
('FLIGHT03', 1200, 'Dammam', 'Riyadh', '2023-12-12', '2023-12-12', 'PILOT03'),  
('FLIGHT04', 1300, 'Riyadh', 'Dubai', '2023-12-13', '2023-12-13', 'PILOT04'),  
('FLIGHT05', 1400, 'Dubai', 'Riyadh', '2023-12-14', '2023-12-14', 'PILOT05'),  
('FLIGHT06', 1500, 'Riyadh', 'Cairo', '2023-12-15', '2023-12-15', 'PILOT06'),
```

```
('FLIGHT07', 1600, 'Jeddah', 'Abu Dhabi', '2023-12-16', '2023-12-16', 'PILOT07'),  
('FLIGHT08', 1700, 'Dammam', 'Medina', '2023-12-17', '2023-12-17', 'PILOT08'),  
('FLIGHT09', 1800, 'Riyadh', 'AlUla', '2023-12-18', '2023-12-18', 'PILOT09'),  
('FLIGHT10', 1900, 'Jeddah', 'Hail', '2023-12-19', '2023-12-19', 'PILOT10');
```

```
INSERT INTO RESERVATION (RESERVATION_ID, DATE_OF_DEPARTURE, DATE_OF_ARRIVAL,  
LUGGAGE_ALLOWANCE, FLIGHT_ID)
```

```
VALUES ('RESERVATION01', '2023-12-10', '2023-12-10', 20, 'FLIGHT01'),  
('RESERVATION02', '2023-12-11', '2023-12-11', 25, 'FLIGHT02'),  
('RESERVATION03', '2023-12-12', '2023-12-12', 30, 'FLIGHT03'),  
('RESERVATION04', '2023-12-13', '2023-12-13', 35, 'FLIGHT04'),  
('RESERVATION05', '2023-12-14', '2023-12-14', 40, 'FLIGHT05'),  
('RESERVATION06', '2023-12-15', '2023-12-15', 45, 'FLIGHT06'),  
('RESERVATION07', '2023-12-16', '2023-12-16', 50, 'FLIGHT07'),  
('RESERVATION08', '2023-12-17', '2023-12-17', 55, 'FLIGHT08'),  
('RESERVATION09', '2023-12-18', '2023-12-18', 60, 'FLIGHT09'),  
('RESERVATION10', '2023-12-19', '2023-12-19', 65, 'FLIGHT10');
```

```
INSERT INTO TICKET (TICKET_ID, RESERVATION_ID, FARE, MEAL)
```

```
VALUES ('TICKET01', 'RESERVATION01', 500, 'Vegetarian'),  
('TICKET02', 'RESERVATION02', 550, 'Standard'),  
('TICKET03', 'RESERVATION03', 600, 'Gluten-free'),  
('TICKET04', 'RESERVATION04', 1500, 'Seafood'),  
('TICKET05', 'RESERVATION05', 1650, 'Vegan'),  
('TICKET06', 'RESERVATION06', 1800, 'Halal'),  
('TICKET07', 'RESERVATION07', 3000, 'Low-Calories'),  
('TICKET08', 'RESERVATION08', 3300, 'Diabetic-friendly'),  
('TICKET09', 'RESERVATION09', 3600, 'Low-carb'),
```

('TICKET10', 'RESERVATION10', 500, 'Standard');

INSERT INTO PASSENGER (PASSENGER_ID, FNAME, LNAME, EMAIL, PHONE_NUM,
NATIONALITY, DOB, GENDER, RESERVATION_ID)

VALUES ('PASSENGER01', 'John', 'Smith', 'john.smith@email.com', '+966 50 1234567',
'American', '1980-01-01', 'Male', 'RESERVATION01'),

('PASSENGER02', 'Jane', 'Doe', 'jane.doe@email.com', '+966 51 7654321', 'British', '1985-02-02',
'Female', 'RESERVATION02'),

('PASSENGER03', 'Ahmed', 'Ali', 'ahmed.ali@email.com', '+966 55 9876543', 'Saudi Arabian',
'1990-03-03', 'Male', 'RESERVATION03'),

('PASSENGER04', 'Fatima', 'Mohammed', 'fatima.mohammed@email.com', '+966 54 3210987',
'Emirati', '1995-04-04', 'Female', 'RESERVATION04'),

('PASSENGER05', 'Omar', 'Abdullah', 'omar.abdullah@email.com', '+966 56 7890123', 'Egyptian',
'2000-05-05', 'Male', 'RESERVATION05'),

('PASSENGER06', 'Sarah', 'Ahmed', 'sarah.ahmed@email.com', '+966 53 1234987', 'Jordanian',
'2005-06-06', 'Female', 'RESERVATION06'),

('PASSENGER07', 'Khalid', 'Hasan', 'khalid.hasan@email.com', '+966 52 9870123', 'Lebanese',
'2010-07-07', 'Male', 'RESERVATION07'),

('PASSENGER08', 'Amira', 'Omar', 'amira.omar@email.com', '+966 57 0123456', 'Turkish', '2015-
08-08', 'Female', 'RESERVATION08'),

('PASSENGER09', 'Hassan', 'Ali', 'hassan.ali@email.com', '+966 58 4567890', 'Moroccan', '2020-
09-09', 'Male', 'RESERVATION09'),

('PASSENGER10', 'Halima', 'Fatima', 'halima.fatima@email.com', '+966 59 8901234', 'Indian',
'2022-10-10', 'Female', 'RESERVATION10');

INSERT INTO CABIN_TYPE (CABIN_TYPE, CABIN_FARE, AIRPLANE_ID)

VALUES ('Economy Class', 500, 'AIRPLANE01'),

('Business Class', 1500, 'AIRPLANE01'),

('First Class', 3000, 'AIRPLANE01'),

```
('Economy Class', 550, 'AIRPLANE02'),  
( 'Business Class', 1650, 'AIRPLANE02'),  
( 'First Class', 3300, 'AIRPLANE02'),  
( 'Economy Class', 600, 'AIRPLANE03'),  
( 'Business Class', 1800, 'AIRPLANE03'),  
( 'First Class', 3600, 'AIRPLANE03');
```

```
INSERT INTO SEAT (SEAT_ID, CABIN_TYPE, AIRPLANE_ID, FLIGHT_ID)  
VALUES ('SEAT01', 'Economy Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT02', 'Economy Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT03', 'Economy Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT211', 'First Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT178', 'Economy Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT179', 'Economy Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT180', 'Economy Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT181', 'Business Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT182', 'Business Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT183', 'Business Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT209', 'Business Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT210', 'Business Class', 'AIRPLANE01', 'FLIGHT01'),  
( 'SEAT212', 'First Class', 'AIRPLANE01', 'FLIGHT01');
```

STEP 9.C) 10 SQL QUERIES

1- List full details of all flights departing from Riyadh on December 12th, 2023.
SELECT * FROM FLIGHT

WHERE DEPARTURE_FROM = 'Riyadh' AND DEPARTURE_DATE = '2023-12-12';

Input

Run SQL

```
SELECT * FROM FLIGHT
WHERE DEPARTURE_FROM = 'Riyadh' AND DEPARTURE_DATE = '2023-12-12';
```

Output

FLIGHT_ID	FLIGHT_FARE	DESTINATION	DEPARTURE_FROM	ARRIVAL_DATE	DEPARTURE_DATE	PILOT_ID
FLIGHT03	1200	Dammam	Riyadh	2023-12-12	2023-12-12	PILOT03

2- Find the total number of passengers traveling on flights departing on December 10th, 2023.

SELECT COUNT(*)

FROM PASSENGER

INNER JOIN RESERVATION ON PASSENGER.RESERVATION_ID = RESERVATION.RESERVATION_ID

INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID

WHERE DEPARTURE_DATE = '2023-12-10';

Input

Run SQL

```
SELECT COUNT(*)
FROM PASSENGER
INNER JOIN RESERVATION ON PASSENGER.RESERVATION_ID = RESERVATION.RESERVATION_ID
INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID
WHERE DEPARTURE_DATE = '2023-12-10';
```

Output

COUNT(*)
1

3- List all the flights operated by pilot Ahmed Ali.

SELECT * FROM FLIGHT

WHERE PILOT_ID = 'PILOT01';

Input

Run SQL

```

SELECT FLIGHT_ID, FLIGHT_FARE, DESTINATION, DEPARTURE_FROM, ARRIVAL_DATE, DEPARTURE_DATE
FROM FLIGHT
INNER JOIN PILOT ON FLIGHT.PILOT_ID = PILOT.PILOT_ID
WHERE PILOT.PFNAME = 'Ahmed' AND PILOT.PLNAME = 'Ali';

```

Output

FLIGHT_ID	FLIGHT_FARE	DESTINATION	DEPARTURE_FROM	ARRIVAL_DATE	DEPARTURE_DATE
FLIGHT01	1000	Riyadh	Jeddah	2023-12-10	2023-12-10

4- Find the average luggage allowance for flights departing from Jeddah in December 2023.

```
SELECT AVG(LUGGAGE_ALLOWANCE)
```

```
FROM RESERVATION
```

```
INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID
```

```
WHERE DEPARTURE_FROM = 'Jeddah' AND DEPARTURE_DATE BETWEEN '2023-12-01' AND '2023-12-31';
```

Input

Run SQL

```

SELECT AVG(LUGGAGE_ALLOWANCE)
FROM RESERVATION
INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID
WHERE DEPARTURE_FROM = 'Jeddah' AND DEPARTURE_DATE BETWEEN '2023-12-01' AND '2023-12-31';

```

Output

AVG(LUGGAGE_ALLOWANCE)
20

5- List all passengers traveling on flights departing on December 15th, 2023, along with their flight details and email addresses.

```
SELECT PASSENGER.FNAME, PASSENGER.LNAME, PASSENGER.EMAIL, FLIGHT.FLIGHT_ID,
FLIGHT.DESTINATION, FLIGHT.DEPARTURE_DATE
```

```
FROM PASSENGER
```

INNER JOIN RESERVATION ON PASSENGER.RESERVATION_ID = RESERVATION.RESERVATION_ID

INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID

WHERE DEPARTURE_DATE = '2023-12-15';

Input

```
SELECT PASSENGER.FNAME, PASSENGER.LNAME, PASSENGER.EMAIL, FLIGHT.FLIGHT_ID, FLIGHT.DESTINATION, FLIGHT.DEPARTURE_DATE
FROM PASSENGER
INNER JOIN RESERVATION ON PASSENGER.RESERVATION_ID = RESERVATION.RESERVATION_ID
INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID
WHERE DEPARTURE_DATE = '2023-12-15';
```

Run SQL

Output

FNAME	LNAME	EMAIL	FLIGHT_ID	DESTINATION	DEPARTURE_DATE
Sarah	Ahmed	sarah.ahmed@email.com	FLIGHT06	Riyadh	2023-12-15

6- Find the total revenue generated from ticket sales for flights departing on December 10th, 2023.

SELECT SUM(FARE)

FROM TICKET

INNER JOIN RESERVATION ON TICKET.RESERVATION_ID = RESERVATION.RESERVATION_ID

INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID

WHERE DEPARTURE_DATE = '2023-12-10';

Input

```
SELECT SUM(FARE)
FROM TICKET
INNER JOIN RESERVATION ON TICKET.RESERVATION_ID = RESERVATION.RESERVATION_ID
INNER JOIN FLIGHT ON RESERVATION.FLIGHT_ID = FLIGHT.FLIGHT_ID
WHERE DEPARTURE_DATE = '2023-12-10';
```

Run SQL

Output

SUM(FARE)
500

7- List all airplanes with a capacity of more than 200 passengers.

SELECT * FROM AIRPLANE

WHERE CAPACITY > 200;

Input

Run SQL

```
SELECT * FROM AIRPLANE
WHERE CAPACITY > 200;
```

Output

AIRPLANE_ID	MODEL_NUMBER	CAPACITY
AIRPLANE03	Airbus A350-900	325
AIRPLANE04	Boeing 787-9	312
AIRPLANE05	Airbus A380-800	544
AIRPLANE06	Boeing 747-8	467

8- Order the used cabin type according to the most used.

```
SELECT CABIN_TYPE, COUNT(*) AS frequency
```

```
FROM SEAT
```

```
GROUP BY CABIN_TYPE
```

```
ORDER BY frequency DESC;
```

Input

Run SQL

```
SELECT CABIN_TYPE, COUNT(*) AS frequency
FROM SEAT
GROUP BY CABIN_TYPE
ORDER BY frequency DESC;
```

Output

CABIN_TYPE	frequency
Economy Class	6
Business Class	5
First Class	2

9- List all pilots who are flying to Dubai in December 2023.

```
SELECT PILOT.PFNAME, PILOT.PLNAME, FLIGHT.DESTINATION
```

```
FROM PILOT
```

```
INNER JOIN FLIGHT ON PILOT.PILOT_ID = FLIGHT.PILOT_ID
```

WHERE DESTINATION = 'Dubai' AND DEPARTURE_DATE BETWEEN '2023-12-01' AND '2023-12-31';

Input

```
SELECT PILOT.PFNAME, PILOT.PLNAME, FLIGHT.DESTINATION
FROM PILOT
INNER JOIN FLIGHT ON PILOT.PILOT_ID = FLIGHT.PILOT_ID
WHERE DESTINATION = 'Dubai' AND DEPARTURE_DATE BETWEEN '2023-12-01' AND '2023-12-31';
```

Run SQL

Output

PFNAME	PLNAME	DESTINATION
Khalid	Hasan	Dubai

10- Find the average flight fare for flights departing from Riyadh to Dammam in December 2023.

SELECT AVG(FLIGHT_FARE)

FROM FLIGHT

WHERE DEPARTURE_FROM = 'Riyadh' AND DESTINATION = 'Dammam' AND DEPARTURE_DATE BETWEEN '2023-12-01' AND '2023-12-31';

Input

```
SELECT AVG(FLIGHT_FARE)
FROM FLIGHT
WHERE DEPARTURE_FROM = 'Riyadh' AND DESTINATION = 'Dammam' AND DEPARTURE_DATE BETWEEN '2023-12-01' AND '2023-12-31';
```

Run SQL

Output

AVG(FLIGHT_FARE)
1200

STEP 10) HOW THE DESIGNED DATABASE SOLVED THE PROBLEM

Our SQL Database performs the general functionality of an airline reservation system with several flights, airplanes, pilots, passengers, tickets, seats, and cabins. Riyadh Airlines announced the operation of one type of airplane; however, our proposed solution can handle

more airplane types which can be easily visualized in our Airplane table. Our goal was to create a system that could help the airline manage its flight reservations and passenger data in a more streamlined and organized way.

We have also ensured that the system is designed to support Riyadh Airline's data management needs, with a normalized ERD and functioning SQL that can execute different queries as needed. Our database allows individual and group reservations, which can help the airline improve customer experience and open more paths for future data analysis.

While our solution may not be as robust or of high quality as some of the more advanced reservation systems in the market, we believe that it provides a solid foundation for Riyadh Airlines to build on and adjust as suitable to entertain reservation needs. We hope that our database design can help the airline improve its operations and better serve its customers.

In conclusion, we realize that the airline's reservations systems are not an easy system to design and the operations and details it includes are not to be restricted to what has been shown in this project. We believe that this is a project that could be expanded widely and with more computing power, it could include so much more information to be saved.